Fraud Detection Metrics Explained

Accuracy • Precision • Recall • F1-Score • ROC-AUC

Real Example: Your Fraud Model tested 1000 credit card transactions.

THE TEST SCENARIO

Out of 1000 test transactions:

- Real frauds: 10 (1%)
- Real legitimate: 990 (99%)

(This is realistic — fraud is rare)

Your model predictions on these 1000 transactions:

- Predicted 50 as fraud
- Predicted 950 as legitimate

STEP 1: THE CONFUSION MATRIX

This is what actually happened:

	Actually Fraud	Actually Legitimate	Total
Predicted Fraud	8 (TP)	42 (FP)	50
Predicted Legitimate	2 (FN)	948 (TN)	950
Total	10	990	1000

Key Terms:

TP (True Positive) = 8: Model said "fraud" and it WAS fraud ✓

FP (False Positive) = 42: Model said "fraud" but it was LEGITIMATE ✗

FN (False Negative) = 2: Model said "legitimate" but it WAS FRAUD ✗✗

TN (True Negative) = 948: Model said "legitimate" and it WAS legitimate ✓

METRIC 1: ACCURACY

Formula:

$$\text{Accuracy} = \frac{TP + TN}{FP + FN + TP + TN}$$

Accuracy = (TP + TN) / Total Accuracy = (8 + 948) / 1000 = 956 / 1000 = 95.6%

What it means:

Out of 1000 predictions, 956 were correct. Your model is 95.6% accurate.

⚠️ BUT WAIT — This is misleading!

What if your model just predicted everything as "LEGITIMATE"? Then: TP=0, FP=0, FN=10, TN=990 Accuracy = (0 + 990) / 1000 = 99% Yet this model catches ZERO fraud! Useless!

❌ Why accuracy fails for fraud detection:

Fraud is rare (1%). A lazy model that says "always legitimate" gets 99% accuracy but catches zero fraud. Accuracy alone doesn't tell you if your model is actually useful.

METRIC 2: PRECISION

Formula:

Precision = TP / (TP + FP) Precision = 8 / (8 + 42) = 8 / 50 = 16%

What it means:

Out of 50 transactions your model flagged as fraud, only 8 were actually fraud.

OR: When your model says "fraud", it's right only 16% of the time.

OR: Out of 50 alerts to your fraud team, 42 are false alarms.

Real-world impact:

Your fraud team reviews 50 alerts. 42 are wasted effort. That's expensive.

✅ When precision matters:

> Loan approval: False positive = reject good customer = lose business. Email spam: False positive = mark real email as spam = user misses important email. Must be careful not to flag legitimate users.

METRIC 3: RECALL (aka Sensitivity)

Formula:

Recall = TP / (TP + FN) Recall = 8 / (8 + 2) = 8 / 10 = 80%

What it means:

Out of 10 real frauds in your test data, your model caught 8 of them.

OR: Your model detects 80% of fraud.

OR: 2 real frauds slipped through undetected.

Real-world impact:

2 frauds go undetected. Customers lose money. Damage to trust.

✅ **Why recall is critical for fraud:**

Missing fraud (false negative) is VERY costly. • Customer loses money • Bank faces chargeback • Customer stops using bank So fraud detection prioritizes recall.

THE TRADEOFF: Precision vs Recall

You can't have both perfect precision AND perfect recall. It's a tradeoff.

Raise detection threshold (flag fewer as fraud):

→ Precision UP (fewer false alarms)

→ Recall DOWN (miss more fraud)

Lower detection threshold (flag more as fraud):

→ Recall UP (catch more fraud)

→ Precision DOWN (more false alarms)

Example with your model:

Current threshold = 0.5

Precision = 16%, Recall = 80%

If you raise threshold to 0.7:

Maybe flag only 20 as fraud (instead of 50)

Catch 8 of them (same 8 as before)

Precision = 8/20 = 40% (better!)

Recall = 8/10 = 80% (same)

METRIC 4: F1-SCORE

Formula:

F1 = 2 × (Precision × Recall) / (Precision + Recall) F1 = 2 × (0.16 × 0.80) / (0.16 + 0.80) F1 = 2 × (0.128) / (0.96) = 0.267 = 26.7%

What it means:

F1 is the harmonic mean of precision and recall. It balances both.

Your F1 is 26.7% — this model is not very good overall.

Why F1 instead of just average?

Harmonic mean penalizes extreme values. If precision=1% and recall=99%, average=50% (misleading). Harmonic mean=1.98% (realistic).

Rule of thumb:

- F1 > 80% = Good model
- F1 50-80% = Decent model
- F1 < 50% = Poor model (needs improvement)
- Your F1 = 26.7% → Model needs improvement

METRIC 5: ROC-AUC

ROC = Receiver Operating Characteristic

AUC = Area Under the Curve

ROC-AUC measures model's ability to distinguish between fraud and legitimate across ALL decision thresholds.

Interpretation:

- AUC = 1.0: Perfect model (always correct)
- AUC = 0.5: Random model (coin flip)
- AUC = 0.0: Worst model (always wrong)

For your model: AUC might be ~0.92

This means: 92% chance your model ranks a random fraud higher than a random legitimate. Very good!

Why ROC-AUC is better than accuracy for imbalanced data:

With 99% legitimate, a model that predicts all legitimate = 99% accuracy but 0% useful. ROC-AUC would give it ~0.5 (random), exposing the truth. ROC-AUC is immune to class imbalance.

SUMMARY TABLE — All Metrics

Metric	Formula	Your Value	Meaning	Good?
Accuracy	(TP+TN)/Total	**95.6%**	Seems great but misleading with imbalanced data	⚠️ Weak
Precision	TP/(TP+FP)	**16%**	Of 50 fraud alerts, 8 correct. 42 false alarms.	❌ Poor
Recall	TP/(TP+FN)	**80%**	Catches 8 of 10 real frauds. Misses 2.	✅ Good
F1-Score	2×P×R/(P+R)	**26.7%**	Balance of precision & recall. Low overall.	❌ Poor
ROC-AUC	0 to 1	**~0.92**	92% chance ranks fraud > legitimate. Excellent!	✅ Great

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WHICH METRIC SHOULD YOU USE?

For Fraud Detection:

✅ Primary: Recall (catch as much fraud as possible) ✅ Secondary: F1-Score (balance precision and recall) ✅ Validation: ROC-AUC (immune to imbalance) ❌ Avoid: Accuracy (misleading with rare fraud)

For Loan Approval:

Precision matters more. False positive = reject good customer = lose business.

For Medical Diagnosis:

Recall matters more. False negative = miss disease = patient dies.

For Email Spam:

Precision matters more. False positive = mark real email as spam = user misses important email.

HOW TO IMPROVE YOUR MODEL

Your current model: Recall=80%, Precision=16%, F1=26.7%

Problem: Too many false alarms (42 out of 50 fraud alerts are wrong)

Option 1: Raise decision threshold

Default threshold = 0.5 probability. Raise to 0.7 or 0.8.

Result: Fewer false alarms, higher precision, same or lower recall.

Trade-off: Might miss a few more frauds but alerts are more reliable.

Option 2: Better features

Create better features (behavioral patterns, anomaly scores, velocity checks).

Result: Model learns better patterns, both precision and recall improve.

Option 3: Handle imbalance better

Use SMOTE more aggressively. Weight fraud cases higher during training.

Result: Model focuses more on minority class (fraud).

Option 4: Use different model

Try LightGBM or CatBoost instead of XGBoost.

Result: Different model might capture patterns better.